

# Hyunchul Park

Ph.D. Student

Cho Chun Shik Graduate School of Mobility

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## Research Interests

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Humanoid Robot Drivers, Human Factors and Human–Machine Interaction in Traffic Safety, Driving Simulator and VR Experiments, Safety Testing of Automated Vehicles, Reinforcement Learning, V2X

## Education

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### Ph.D. Student

2025 – Present

Cho Chun Shik Graduate School of Mobility, KAIST

Research: closing the embodiment gap for humanoid robot drivers

Advisor: Inhi Kim, Ph.D.

### M.S.

2023 – 2025

Cho Chun Shik Graduate School of Mobility, KAIST

Research: driver human factors in driving-simulator and VR experiments

Advisor: Inhi Kim, Ph.D.

### B.S. in Civil & Traffic Engineering

2016 – 2022

Kongju National University

## Publications

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### Articles in International Journals

1. Baek, J., Kim, J., **Park, H.**, & Kim, I. (2026). How age-related constraints reshape transport disadvantage in dynamic railway information systems: Evidence from virtual aging simulation. *Journal of Urban Mobility*, 10, 100292. <https://doi.org/10.1016/j.urbmob.2026.100292>
2. **Park, H.**, Oh, T., Lim, J., & Kim, I. (2026). Micro mobility safety challenges: A study on drivers overtaking bicycles and e-scooters according to road conditions and prior riding experience. *Transportation Research Part F: Traffic Psychology and Behaviour*, 117, 103482. <https://doi.org/10.1016/j.trf.2025.103482>
3. Lim, J., **Park, H.**, Oh, T., & Kim, I. (2025). The impact of information delivery systems in tunnels depending on lighting intensity and speed limit. *Tunnelling and Underground Space Technology*, 157, 106365. <https://doi.org/10.1016/j.tust.2025.106365>

4. **Park, H.**, Oh, T., & Kim, I. (2024). Effects of driver's braking behavior by the real-time pedestrian scale warning system. *Accident Analysis & Prevention*, 205, 107685. <https://doi.org/10.1016/j.aap.2024.107685>

### ***Articles in Domestic Journals (in Korean)***

1. **Park, H.**, Oh, T., & Kim, I. (2022). A study on validation of omnidirectional VR treadmill by comparison of spatial orientation skills. *The Journal of The Korea Institute of Intelligent Transport Systems*, 21(5), 15–27.

### ***Articles Submitted***

1. **Park, H.**, Oh, T., Liu, Y., & Kim, I. (2026). Fair adversarial testing of automated vehicles in highway cut-ins: Reference-shielded attacks and a capability-controlled evaluation. *IEEE Transactions on Intelligent Vehicles* (under review).
2. **Park, H.**, Oh, T., Kim, J., Kim, J., & Kim, I. (2026). From passive visibility to active permissibility: A virtual reality study on overcoming track avoidance bias in subway station fires. *Underground Space* (under 1<sup>st</sup> revision).
3. Kim, J., Kim, M., Kim, J., **Park, H.**, & Kim, I. (2026). Risks of static obstacle placement in crowd evacuation: Density sensitivity and the isolation of beneficial zones. (under review).

### ***Manuscripts in Preparation***

1. **Park, H.**, & Kim, I. When the car moves the driver: Body loops and the speed limit of a simulated humanoid driver.
2. **Park, H.**, & Kim, I. Seeing from a swaying body: Designing head-camera compensation for humanoid robot drivers from off-vehicle tests.
3. **Park, H.**, & Kim, I. What an end-to-end driving brain must know about a humanoid body: Closed-loop evaluation with an identified robot driver.
4. **Park, H.**, & Kim, I. Roadside information, yellow intervals and crossing capacity for humanoid robot drivers: A matched-safety simulation at an occluded intersection.
5. **Park, H.**, Tamakloe, R., Kim, E. M., Lim, J., & Kim, I. Human-centric evaluation of UWB-based V2P ADAS: How pedestrian tag adoption rates influence driver cognitive load and safety performance in school zones.

### **Conferences**

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#### ***Articles in International Conference Proceedings***

1. **Park, H.**, & Kim, I. (2026, July). Severity-controllable adversarial testing of highway automated vehicles using naturalistic aggressive-moment statistics. In *15th Asia-Pacific Conference on Transportation and the Environment (APTE)*, Jeju, Korea. Oral presentation.
2. **Park, H.**, Oh, T., Lim, J., & Kim, I. (2025). Micromobility safety challenges: A study on drivers overtaking bicycles and e-scooters according to road conditions and cross-modal experience. In *Transportation Research Board 104th Annual Meeting*, Washington, DC. Oral presentation; **TRB Outstanding Paper Award**.

3. **Park, H.**, Oh, T., Lim, J., & Kim, I. (2024). Micro mobility safety challenges: A study on drivers overtaking bicycles and e-scooters in relation to road conditions and prior riding experience. In *36th ICTCT Conference*, The Hague, Netherlands. Oral presentation.
4. Lim, J., **Park, H.**, Oh, T., & Kim, I. (2024). The impact of information delivery systems in tunnels depending on lighting intensity and speed limit. In *Transportation Research Board 103rd Annual Meeting*, Washington, DC. Poster.
5. **Park, H.**, Oh, T., & Kim, I. (2023). Effects of driver's braking response by the real-time pedestrian scale warning system. In *Transportation Research Board 102nd Annual Meeting*, Washington, DC. Poster.
6. **Park, H.**, Oh, T., & Kim, I. (2022). Validation of spatial orientation skill using virtual reality treadmill. In *ITS World Congress 2022*, Los Angeles, CA. Oral presentation.

## Research Grants and Projects

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### Current Projects

1. **World-Model-Based Driving Skill Learning for Humanoid Robots and Real-Vehicle Validation in a V2X Testbed** 2026 – 2028  
*National Research Foundation of Korea (NRF), Doctoral Student Research Encouragement Program*  
 Role: Principal Investigator, Amount: KRW 50,000,000
2. **Development of Virtual Environment and Demonstration Technology for Automated Driving based on Metaverse** 2023 – 2027  
*Ministry of Land, Infrastructure and Transport (MOLIT)*  
 Role: Researcher (behaviorally realistic traffic agents for automated-driving tests)

### Selected Completed Projects

1. **Developing an Aggressive Pedestrian Model for Autonomous Vehicle Safety Testing** 2024 – 2025  
*National Research Foundation of Korea (NRF), Graduate Student Research Encouragement Fund*  
 Role: Principal Investigator, Amount: KRW 12,000,000
2. **ASEAN–Korea Transport Cooperation Roadmap (2026–2030)** 2024 – 2025  
*Ministry of Land, Infrastructure and Transport (MOLIT), with the ASEAN Secretariat*  
 Role: Planning lead; presented at the ASEAN Land Transport Working Group, Cambodia, Aug. 2025

## Awards and Honors

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| 2026 | National Excellence Scholarship (Science & Engineering), <i>Korea Student Aid Foundation</i> |
| 2025 | Outstanding Paper Award, <i>Transportation Research Board 104th Annual Meeting</i>           |
| 2025 | Young Investigator Award, <i>IPESK</i>                                                       |
| 2024 | Graduate Student Outstanding Paper Award, <i>KAIST</i>                                       |
| 2024 | Young-Han Kim Global Leader Scholarship, <i>KAIST</i> (25 students annually)                 |

- 2024 Best Teaching Assistant Award, *KAIST*
- 2024 President's Award (Distinguished Graduate), *Korean Society of Transportation*
- 2023 Best Paper Award, *The Korea Institute of Intelligent Transport Systems*
- 2021 Best Paper Award, *The Korea Institute of Intelligent Transport Systems*
- 2021 Excellence Award, Marine Big Data Analysis Competition, *UNIST*
- 2020 President's Award (Capstone Design), *Kongju National University*

## Talks

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1. "Building an LLM-Powered Wiki for Researchers: Toward Automated Knowledge Curation in Research Workflows," AI Agent Seminar, Korea Transport Institute (KOTI), Jun. 2026.

## Teaching

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| <b>Teaching Assistant</b><br>Experiments in Future Mobility within the Metaverse, KAIST<br>Best Teaching Assistant Award | 2024 |
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*Last updated: October 2026*